Mathematics: analysis and approaches
Standard level
Paper 2
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 2 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P2–02

Student name

30 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- A graphic display calculator is required for this paper.
- This paper has a Section A only. Answer all questions. Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[26 marks]**.

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

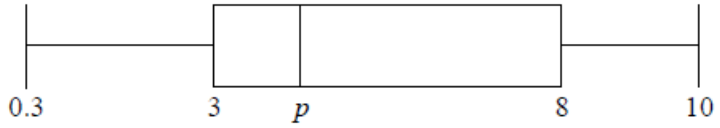
1. [Maximum mark: 5]

Ten students were asked for the distance, in km, from their home to school. Their responses are recorded below.

0.3 0.4 3 3 3.5 5 7 8 8 10

(a) For these data, find the mean distance from a student's home to school. [2]

The following box-and-whisker plot represents this data.



(b) Find the value of p. [1]

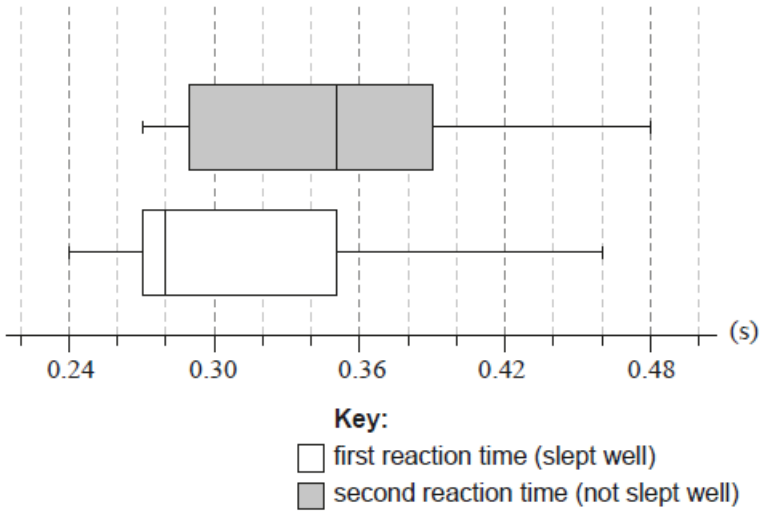
(c) Find the interquartile range. [2]

2. [Maximum mark: 6]

A random sample of nine adults were selected to see whether sleeping well affected their reaction times to a visual stimulus. Each adult's reaction time was measured twice.

The first measurement for reaction time was taken on a morning after the adult had slept well. The second measurement was taken on a morning after the same adult had not slept well.

The box and whisker diagrams for the reaction times, measured in seconds, are shown below.



Consider the box and whisker diagram representing the reaction times after sleeping well.

(a) State the median reaction time after sleeping well. [1]

(b) Verify that the measurement of 0.46 seconds is not an outlier. [3]

(c) State why it appears that the mean reaction time is greater than the median reaction time. [1]

(d) Now consider the two box and whisker diagrams.

Comment on whether these box and whisker diagrams provide any evidence that might suggest that not sleeping well causes an increase in reaction time. [1]

(This question continues on the following page)

(Question 2 continued)

3. [Maximum mark: 5]

The following table shows the Mathematics test scores (x) and the Science test scores (y) for a group of eight students.

Mathematics scores (x)	64	68	72	75	80	82	85	86
Science scores (y)	67	72	77	76	84	83	89	91

The regression line of y on x for this data can be written in the form $y = ax + b$.

(a) Find the value of a and the value of b. [2]

(b) Write down the value of the Pearson's product-moment correlation coefficient, r. [1]

(c) Use the equation of your regression line to predict the Science test score for a student who has a score of 78 on the Mathematics test. Express your answer to the nearest integer. [2]

(This question continues on the following page)

(Question 3 continued)

4. [Maximum mark: 4]

Let A and B be two independent events such that $P(A \cap B') = 0.16$ and $P(A' \cap B) = 0.36$.

Given that $P(A \cap B) = x$, find the value of x.

5. [Maximum mark: 6]

Events A and B are independent and $P(A) = 3P(B)$.

Given that $P(A \cup B) = 0.68$, find $P(B)$.

(This question continues on the following page)

(Question 5 continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability) and cover subtopics 4.1 to 4.6. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	19M.2.SL.TZ1.S_1 (former Mathematics SL course) · 4.3 mean · 4.2 box-and-whisker plot, interquartile range
Section A — question 2	22M.2.SL.TZ2.5 · 4.2, 4.3 box and whisker diagrams, outliers, skewness
Section A — question 3	22N.2.SL.TZ0.1 · 4.4 regression line of y on x, correlation coefficient r
Section A — question 4	22M.2.SL.TZ1.6, part (a) · 4.6 independent events — part (b), P(A′ | B′) with the formal definition of independence (4.11), left out
Section A — question 5	22M.2.SL.TZ2.4 · 4.6 independent events, combined events

Please **do not** write on this page.

Answers written on this page
will not be marked.